

Администрирование локальных сетей

Статическая маршрутизация VLAN

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Цели и задачи работы

Настроить статическую маршрутизацию VLAN в сети и изучить передачу ICMP-пакетов между устройствами из разных VLAN.

Построение схемы сети

Добавление маршрутизатора Cisco 2811

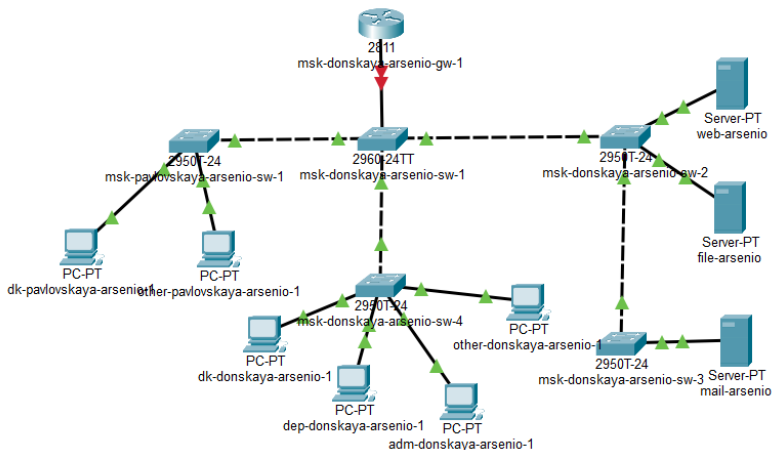


Рис. 1: Размещение маршрутизатора Cisco 2811 в логической области проекта

Первоначальная настройка маршрутизатора

Настройка имени, паролей и SSH

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-donskaya-arsenio-gw-1
msk-donskaya-arsenio-gw-1(config)#line vty 0 4
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#line cons 0
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-gw-1(config)#serv pass
msk-donskaya-arsenio-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-arsenio-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-gw-1(config)#line vty 0 4
*Mar 1 0:2:7.426: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:2:7.426: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-gw-1(config-line)#trans in ssh
msk-donskaya-arsenio-gw-1(config-line)#exit
msk-donskaya-arsenio-gw-1(config)#int f0/0
msk-donskaya-arsenio-gw-1(config-if)#no sh

msk-donskaya-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

Настройка VLAN-маршрутизации

Виртуальные интерфейсы маршрутизатора

```
mek-donskaya-arsenic-gw-1(config-if)#int f0/0.2
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 2
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.1.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr management
mek-donskaya-arsenic-gw-1(config-subif)#int f0/0.3
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 3
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.0.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr servers
mek-donskaya-arsenic-gw-1(config-subif)#int f0/0.101
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.101, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 101
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.3.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr dk
mek-donskaya-arsenic-gw-1(config-subif)#int f0/0.102
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 102
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.4.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr departaments
mek-donskaya-arsenic-gw-1(config-subif)#int f0/0.103
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.103, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 103
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.5.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr adm
mek-donskaya-arsenic-gw-1(config-subif)#int f0/0.104
mek-donskaya-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.104, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

mek-donskaya-arsenic-gw-1(config-subif)#enc dot1q 104
mek-donskaya-arsenic-gw-1(config-subif)#ip addr 10.128.6.1 255.255.255.0
mek-donskaya-arsenic-gw-1(config-subif)#descr arher
```

Проверка связи

Проверка доступности устройств

```
dk-donskaya-arsenio-1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>
C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.10

Pinging 10.128.5.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time=10ms TTL=127

Ping statistics for 10.128.5.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 10.128.6.10

Pinging 10.128.6.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.10: bytes=32 time<1ms TTL=127
Reply from 10.128.6.10: bytes=32 time<1ms TTL=127
Reply from 10.128.6.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Режим Simulation

Передача ICMP-пакета

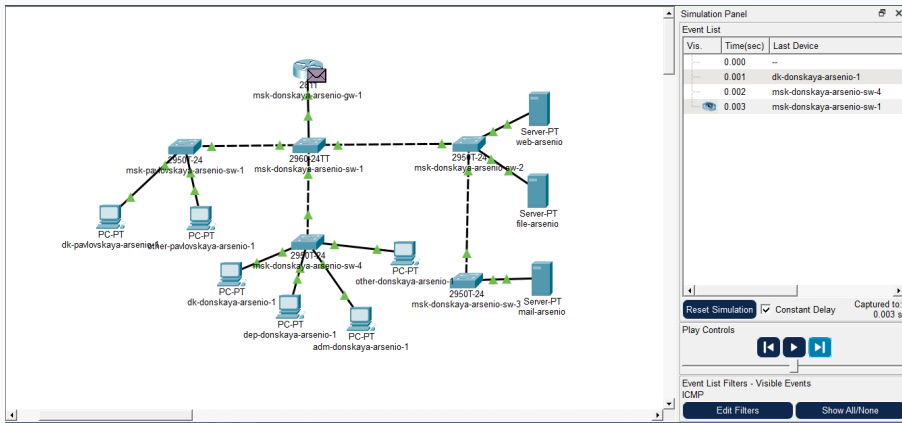


Рис. 5: Передача ICMP-пакета в режиме Simulation

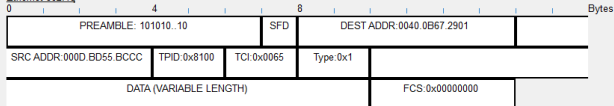
Входящий PDU на маршрутизаторе

PDU Information at Device: msk-donskaya-arsenio-gw-1

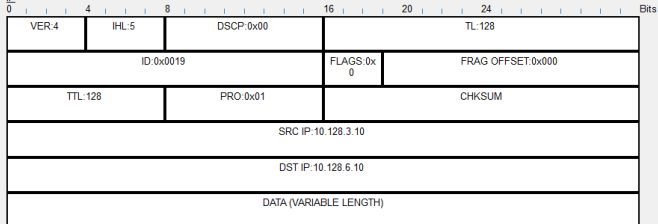
OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

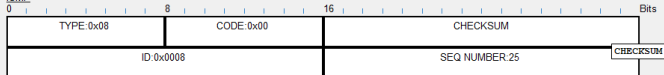
Ethernet 802.1q



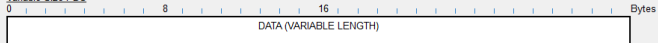
IP



ICMP



Variable Size PDU



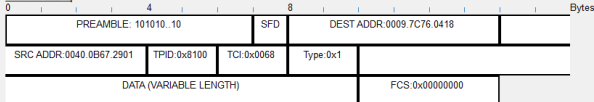
Исходящий PDU на маршрутизаторе

PDU Information at Device: msk-donskaya-arsenio-gw-1

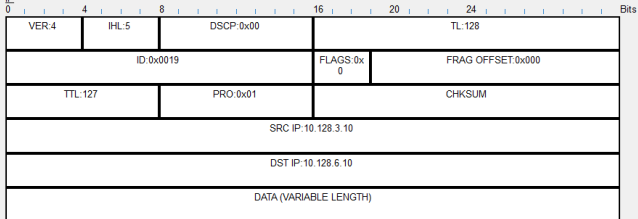
OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

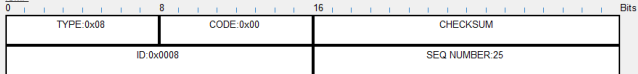
Ethernet 802.1q



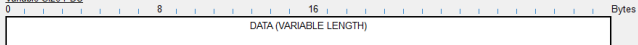
IP



ICMP



Variable Size PDU



Доставка ICMP-пакета

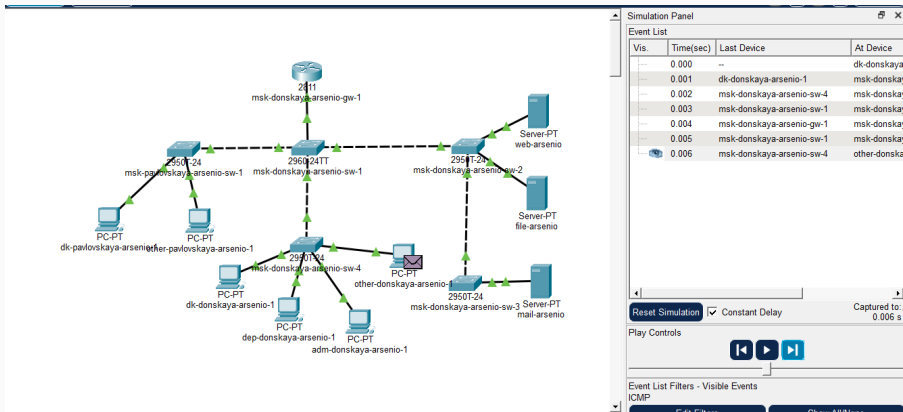


Рис. 8: Доставка ICMP-пакета к конечному устройству

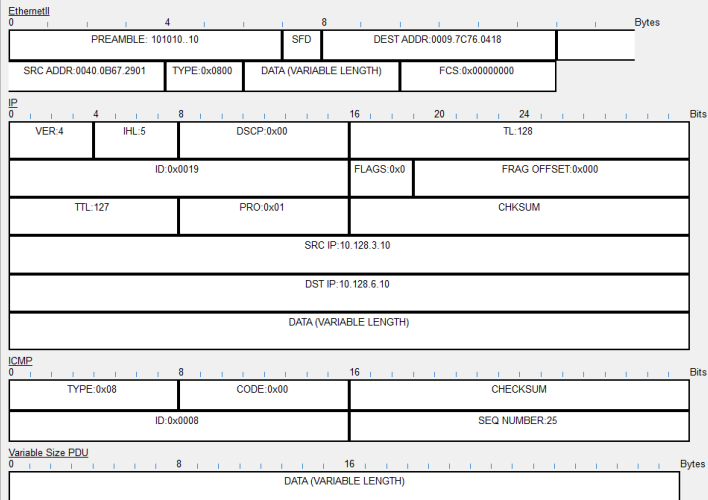
Анализ PDU

Входящий ICMP Echo Request

PDU Information at Device: other-donskaya-arsenio-1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

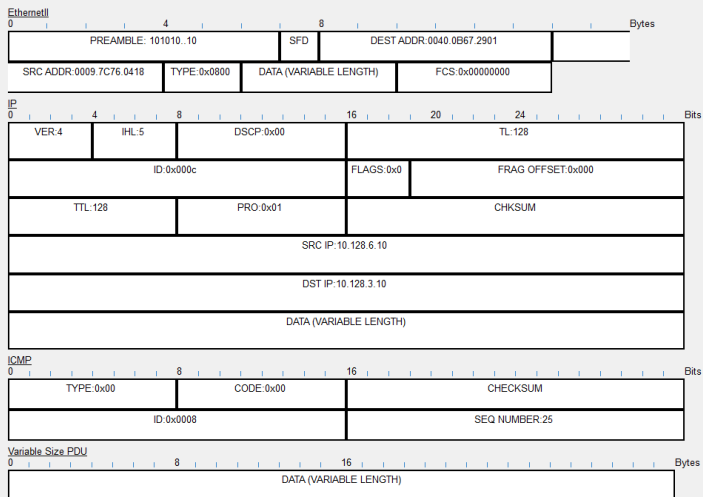


Формирование ICMP Echo Reply

PDU Information at Device: other-donskaya-arsenio-1

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats



Возврат ICMP-пакета

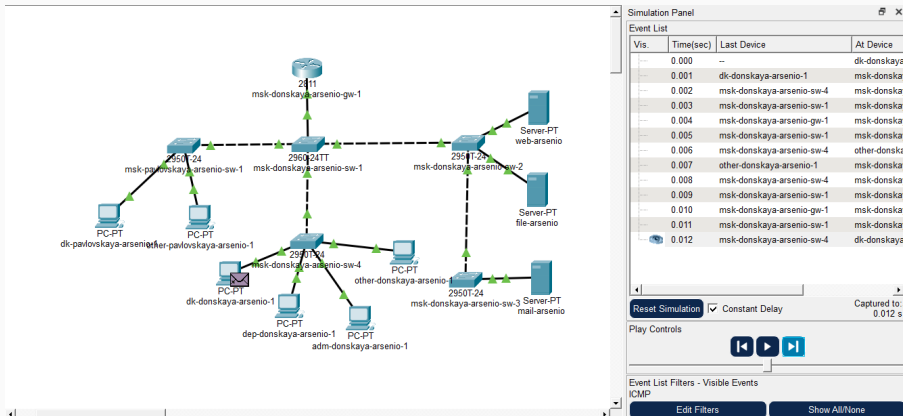


Рис. 11: Возврат ICMP-пакета к исходному устройству

Выводы

В ходе лабораторной работы был добавлен маршрутизатор Cisco 2811 и настроена маршрутизация между VLAN по схеме router-on-a-stick.

На коммутаторе был настроен trunk-порт, а на маршрутизаторе были созданы подинтерфейсы с инкапсуляцией 802.1Q.

Проверка ping подтвердила доступность устройств из разных VLAN. В режиме Simulation был изучен процесс прохождения ICMP-пакетов и изменение заголовков при маршрутизации.